$d=1$ <u>Friedel oscillations at finite T:</u>

1d fermion gas with hard wall at $x=0 \Rightarrow \psi(0)=0$.

$\Rightarrow G(x=0,x') = G(x,x'=0) = 0$ for any kind of GF (R, A, causal or Matsubara).

Consider $G_M(\tau,x,x') = -\langle T_\tau \psi(x,0)\psi^+(x',\tau)\rangle =$

$$= -\theta(\tau)\langle\psi(x,0)\psi^+(x',\tau)\rangle + \theta(-\tau)\langle\psi^+(x',\tau)\psi(x,0)\rangle.$$

Clearly $n(x) = (2)\lim_{\tau\to-0} G_M(\tau,x,x'=x)$

$\hookrightarrow$ spin degeneracy.

GF of the system with hard wall potential

$$G_M^{(1)}(\tau,x,x') = G_M^{(0)}(\tau,x,x') - G_M^{(0)}(\tau,x,-x')$$

where $G_M^{(0)}(\tau,x,x')$ is GF of the gas <u>without</u> the wall.

* Why $G_M^{(1)}$ is the correct GF? $\leftarrow$ it satisfies BC $G_M^{(1)}(\tau,x=0,x') = G_M^{(1)}(\tau,x,x'=0) = 0$:

indeed (1) $G_M^{(1)}(\tau,x,x'=0) = G_M^{(0)}(\tau,x,0) - G_M^{(0)}(\tau,x,0) = 0$ trivially;

(2) $G_M^{(1)}(\tau,x=0,x') = G_M^{(0)}(\tau,0,x') - G_M^{(0)}(\tau,0,-x')$

But we know that $G_M^{(0)}(\tau,x,y)$ is function of the difference $(x-y)$ only [because G_0 is GF of translationally invariant system].

In addition, it can only be function of $|x-y|$, because $G^{(1)}$ is GF of a system invariant under inversion $\underline{x\to-x}$.

Thus, $n(x) = 2\lim_{\tau\to-0} G_M^{(1)}(\tau,x,x'=x) = 2\lim_{\tau\to-0}\left(G_M^{(0)}(\tau,x,x) - G_M^{(0)}(\tau,x,-x)\right).$

Find $G_M^{(0)}$

Calculate $G_R(t,x,y) = -i\,\theta(t) \langle \{ \psi(x,t), \psi^+(y,0) \} \rangle$

(here we deal with fermions, $\varepsilon = +1$, $\Rightarrow [\,,\,]_\varepsilon = \{\,,\,\}$)

Using <u>lect 6</u> $\langle \psi(x,t)\, \psi^+(y,0) \rangle = \frac{1}{V} \sum_{kk'} e^{ikx - i\xi_k t - ik'y}$

$$\cdot \underbrace{\langle \hat{a}_k \hat{a}^+_{k'} \rangle}_{\delta_{kk'}(1-f_k)} = \frac{1}{V} \sum_k e^{ik(x-y) - i\xi_k t} (1-f_k)$$

, where $f_k \equiv \langle \hat{a}^+_k \hat{a}_k \rangle$ (evaluated at finite T!)

Also, $$\langle \psi^+(y,0)\, \psi(x,t) \rangle = \frac{1}{V} \sum_{kk'} e^{ikx - i\xi_k t - ik'y} \underbrace{\langle \hat{a}^+_{k'} \hat{a}_k \rangle}_{\delta_{kk'} f_k}$$

$$= \frac{1}{V} \sum_k e^{ik(x-y) - i\xi_k t} f_k .$$

$$\Rightarrow G_R(t,x,y) = -i\,\theta(t) \langle \psi(x,t)\, \psi^+(y,0) + \psi^+(y,0)\, \psi(x,t) \rangle$$

$$= -i\,\theta(t) \frac{1}{V} \sum_k e^{ik(x-y) - i\xi_k t} \underbrace{(1 - f_k + f_k)}_{1} \quad !$$

$$\Rightarrow G_R^{(0)}(\vec{p}, \omega) = \int d\vec{x} \int_{-\infty}^{\infty} dt\; G_R(t,x,y)\, e^{-i\vec{p}\cdot(\vec{x}-\vec{y}) + i\omega t} =$$

$$= -\frac{i}{V} \sum_k \int dx\, e^{ikx - ipx} \int_0^\infty dt\, e^{i\omega t - i\xi_k t}\, \underline{\underline{e^{-\delta t}}}$$

$$= -\frac{i}{V} \sum_k V \delta_{\vec{k},\vec{p}} \frac{i}{\omega - \xi_k + i\delta} = \frac{1}{\omega - \xi_p + i\delta}$$

<u>Now</u> To obtain Matsubara GF we use $G_M(\omega_n) = G_R(\omega = i\omega_n)$ for $\omega_n > 0$

$$\Rightarrow G_M^{(0)}(\vec{p}, \omega_n) = G_R(\vec{p}, \omega = i\omega_n) =$$

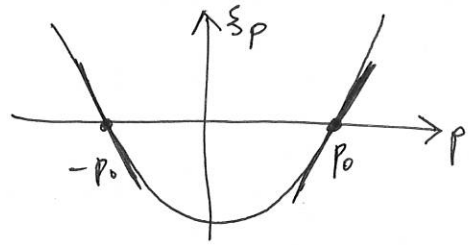
$$= \frac{1}{i\omega_n - \xi_p} \quad \text{(That's all!)}$$

Now F.T. to real space

$$G_M^{(0)}(\tau, r = x-y) = \frac{1}{\beta}\sum_{\omega_n}\int_{-\infty}^{\infty}\frac{dp}{2\pi}\,\frac{e^{ipr}\,e^{-i\omega_n\tau}}{i\omega_n - \xi_p}$$

$$\xi_p = \frac{p^2-p_0^2}{2m} = \frac{(p-p_0)(p+p_0)}{2m} = \begin{cases} v_F K & \text{for } p \approx p_0 + K \\ -v_F K & \text{for } p \approx -p_0 + K \end{cases}, \quad v_F \equiv \frac{p_0}{m}$$

p_0 = Fermi momentum



$$I(x) = \int_{-\infty}^{\infty}\frac{dp}{2\pi}\,\frac{e^{ipx}}{i\omega_n - \xi_p} = \int_{-\infty}^{0}\frac{dp}{2\pi}\,\frac{e^{ipx}}{i\omega_n-\xi_p} + \int_{0}^{\infty}\frac{dp}{2\pi}\,\frac{e^{ipx}}{i\omega_n-\xi_p}$$

$p = -p_0 + K$, $\xi_p = -v_F K$ $\qquad$ $p = p_0 + K$, $\xi_p = v_F K$

$$= \int_{-\infty}^{p_0}\frac{dk}{2\pi}\,\frac{e^{-ip_0x}\,e^{ikx}}{i\omega_n + v_F K} + \int_{-p_0}^{\infty}\frac{dk}{2\pi}\,\frac{e^{ip_0x}\,e^{ikx}}{i\omega_n - v_F K}$$

Let now $\xi = v_F K$, $dk = \frac{d\xi}{v_F}$ and $\int_{-\infty}^{p_0} dk \ldots = \int_{-\infty}^{v_F p_0}\frac{d\xi}{v_F}\ldots$

But $v_F p_0 = \frac{p_0^2}{m} = 2E_F$ which is huge energy

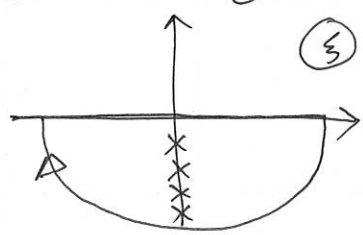
$\Rightarrow$ can set $v_F p_0 \to \infty$. Then

$$I(x) = \int_{-\infty}^{\infty}\frac{d\xi}{2\pi v_F}\,e^{i\xi x/v_F}\left(\frac{e^{-ip_0x}}{i\omega_n+\xi} + \underbrace{\frac{e^{ip_0x}}{i\omega_n-\xi}}_{\text{can change } \xi\to-\xi \text{ here}}\right)$$

$$G^{(0)}_{M}(\tau,x) = \frac{1}{\beta}\sum_{\omega_n} e^{-i\omega_n\tau}\int_{-\infty}^{\infty}\frac{d\xi}{2\pi v_F}\;\frac{e^{ip_0x}\,e^{-i\xi x/v_F} + e^{i\xi x/v_F}\,e^{-ip_0x}}{i\omega_n+\xi}$$

For $x>0$, consider $J_1(x) = \sum_{\omega_n}\int_{-\infty}^{\infty} d\xi\;\frac{e^{-i\xi x/v_F}}{\xi+i\omega_n}$

$$e^{-i\frac{x}{v_F}\xi} = e^{-i\frac{x}{v_F}(\xi'+i\xi'')} = e^{-i\frac{x}{v_F}\xi'}\,e^{\frac{x}{v_F}\xi''} \Rightarrow \text{need } \xi''<0$$



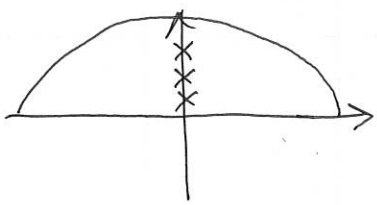
$$J_1(x>0) = \sum_{\omega_n}(-2\pi i)\,e^{-i\frac{x}{v_F}(-i\omega_n)}$$

Residues at $\xi=-i\omega_n$ for $\omega_n>0$

$$J_1(x>0) = -2\pi i\sum_{\omega_n}\theta(\omega_n)\,e^{-\frac{x\omega_n}{v_F}}$$

$$J_2(x>0) = \sum_{\omega_n}\int_{-\infty}^{\infty} d\xi\;\frac{e^{i\frac{x}{v_F}\xi}}{\xi+i\omega_n} = \sum_{\omega_n} 2\pi i\,e^{i\frac{x}{v_F}(-i\omega_n)}\,\theta(-\omega_n)$$

$$e^{i\frac{x}{v_F}(\xi'+i\xi'')} = e^{i\frac{x}{v_F}\xi'}\,e^{-\frac{x}{v_F}\xi''} \Rightarrow \text{need } \xi''>0$$



$$\xi = -i\omega_n > 0 \Rightarrow \omega_n<0$$

$$J_2(x>0) = 2\pi i\sum_{\omega_n}\theta(-\omega_n)\,e^{\frac{x\omega_n}{v_F}} = (\omega_n=-\omega_n')$$

$$= 2\pi i\sum_{\omega_n'}\theta(\omega_n')\,e^{-\frac{x\omega_n'}{v_F}}$$

$$\Rightarrow G^{(0)}_{M}(x,\tau) = \frac{1}{2\pi v_F\beta}\sum_{\omega_n}\Big\{e^{-i\omega_n\tau}\,\theta(\omega_n)\,e^{-\frac{x\omega_n}{v_F}}\,e^{ip_0x}(-2\pi i)$$

$$+\; e^{i\omega_n\tau}\,\theta(\omega_n)\,e^{-\frac{x\omega_n}{v_F}}\,e^{-ip_0x}(2\pi i)\Big\}$$

Observe that $\sum_{\omega_n>0} e^{-i\omega_n\tau - \frac{x}{v_F}\omega_n} = \left(\omega_n = \frac{\pi}{\beta}(2n+1),\ \beta = 1/T\right) =$

$$= \sum_{n=0}^{\infty} e^{-\frac{\pi}{\beta}\left(\frac{x}{v_F}+i\tau\right)(1+2n)} = \left(\text{let } z = e^{-\frac{\pi T}{v_F}(x+iv_F\tau)}\right)$$

$$= z\sum_{n=0}^{\infty} z^{2n} = z\left(1+z^2+z^4+\dots\right) = \frac{z}{1-z^2} =$$

$$= \frac{1}{\frac{1}{z}-z} = \frac{1}{e^{\frac{\pi T}{v_F}(x+i\tau v_F)} - e^{-\frac{\pi T}{v_F}(x+iv_F\tau)}} = \frac{1}{2\,\mathrm{sh}\left[\frac{\pi T}{v_F}(x+i\tau v_F)\right]}$$

And hence $\sum_{\omega_n>0} e^{i\omega_n\tau} e^{-\frac{x\omega_n}{v_F}} = \dfrac{1}{2\,\mathrm{sh}\left[\frac{\pi T}{v_F}(x-iv_F\tau)\right]}$

So that

$$G_M^{(0)}(x,\tau) = \frac{-2\pi i}{2\pi v_F\beta}\left(\frac{e^{ip_0x}}{2\,\mathrm{sh}\,\frac{\pi T}{v_F}(x+iv_F\tau)} - \frac{e^{-ip_0x}}{2\,\mathrm{sh}\,\frac{\pi T}{v_F}(x-iv_F\tau)}\right)$$

We need $\lim_{\tau\to-0} G_M^{(0)}(\tau,x) =$

$$= \frac{-iT}{2v_F}\,\frac{e^{ip_0x}-e^{-ip_0x}}{\mathrm{sh}\left(\frac{\pi Tx}{v_F}\right)} = \frac{T\,\mathrm{sin}(p_0x)}{v_F\,\mathrm{sh}\left(\frac{\pi Tx}{v_F}\right)} \xrightarrow[x\to0]{} \frac{p_0}{\pi}$$

Hence $n(x) = 2\left[G_M^{(0)}(\tau=-0, x=0) - G_M^{(0)}(\tau=-0, 2x)\right] =$

relative distance $r = x-y$

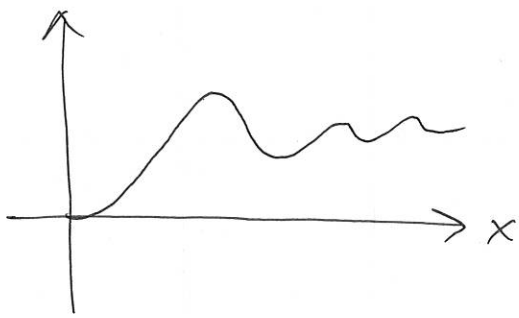
$$= 2\left(\frac{p_0}{\pi} - \frac{T\,\mathrm{sin}(2p_0x)}{v_F\,\mathrm{sh}\left(\frac{2\pi Tx}{v_F}\right)}\right) = \underbrace{\left(\frac{2p_0}{\pi}\right)}_{n_0}\left[1 - \frac{\pi T\,\mathrm{sin}(2p_0x)}{v_F p_0\,\mathrm{sh}\left(\frac{2\pi Tx}{v_F}\right)}\right]$$

n_0 = uniform density far from the wall.

$$n(x) = n_0 \left(1 - \frac{\pi T \, \sin(2p_0 x)}{v_F p_0 \, \mathrm{sh}\left(\frac{2\pi T}{v_F} x\right)} \right)$$

Check: $n(x=0) = 0$ (as it should be, $\psi(x=0)=0$)

<u>$T=0$</u> limit: $$n(x) = n_0 \left(1 - \frac{\sin(2p_0 x)}{2p_0 x} \right)$$



Friedel oscillations with period $\sim \frac{1}{p_0}$.

<u>Consequence of Fermi Surface!</u>

<u>$T \neq 0$</u> $$n(x) \to n_0 \left(1 - \frac{\pi T}{v_F p_0} \sin(2p_0 x) \, e^{-\frac{2\pi T}{v_F} x} \right)$$

<u>for $x \gtrsim \xi$</u>

oscillations are getting damped beyond the thermal scale $\xi \sim \frac{v_F}{T}$ (de Broigle wavelenth)